

Minseong Cho

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🏠 Softlab, Ajou University
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Research Interests

Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

Education

- **M.S. in Mechanical Engineering** Sep 2025 – Present
Ajou University, Suwon, South Korea
◦ **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025
Ajou University, Suwon, South Korea

Research Experience

- **Graduate Research Student – Softlab, Ajou University** Sep 2025 – Present
 - **Mechanics of Moisture-Driven Hair Frizz**
Keywords: Hair mechanics, Moisture-driven frizz
Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung
 - **Boundary-Driven Active Matter in Soft Hydrogels**
Keywords: Active matter, Elastic Leidenfrost effect
Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student – Softlab, Ajou University** Mar 2023 – Aug 2025
 - **Boundary-Driven Active Matter in Soft Hydrogels**
Keywords: Active matter, Elastic Leidenfrost effect
Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung

Conferences and Presentations

- **APS Division of Fluid Dynamics Meeting** (Houston, TX, USA) Nov 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **Asian Congress of Fluid Mechanics** (Seoul, South Korea) Sep 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **KSME Annual Conference** (Jeju, South Korea) Nov 2024
 - Explosive motion of spherical hydrogel via the Leidenfrost effect

Awards & Honors

- **Dean's List – Ajou University** Spring 2024, Spring 2025
Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition – Ajou University** June 2025
2nd Place · KRW 1,000,000 prize
Project: Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025

3rd Place · KRW 300,000 prize

Awarded by: Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)

Project: Compliant-mechanism-based knee assistive device for older adults.

- **APS DFD Travel/Enabling Grant**

2025

US\$750 travel cost reimbursement

Awarded by: APS Division of Fluid Dynamics, American Physical Society

Purpose: Travel support for participation in the APS Division of Fluid Dynamics Meeting.

- **Gallery of Fluid Motion**

Apr 2026

Haechon Choi GFM Award · KRW 400,000 prize

Awarded by: The Korean Society of Mechanical Engineers, Fluid Engineering Division

Title: Explosive motion of spherical hydrogel via the Leidenfrost effect

Authors: Minseong Cho, Yeonsu Jung, Jonghyun Ha

Publications

- **Boundary-driven active matter and stochastic rectification via Leidenfrost impulses**

Minseong Cho, Yeonsu Jung, and Jonghyun Ha

☒ In preparation

- **Poroelectric swelling time as a quantitative marker of hair damage and treatment**

Jaehyuk Cho, Minseong Cho, Selin Cho, Haesoo Park, Hyun-Na Kim, and Jonghyun Ha

☒ In preparation

References

Prof. Jonghyun Ha

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Dr. Yeonsu Jung

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